Separable Differential Equations

ANSWER KEY

Solving by separation of variables

1 Solve the differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 3$.
$y\,dy = x\,dx$, so $\frac{y^2}{2} = \frac{x^2}{2} + C$. Using $y(0) = 3$: $C = \frac{9}{2}$. Solution: $y^2 = x^2 + 9$, i.e. $y = \sqrt{x^2 + 9}$ (taking the positive root since $y(0) = 3 > 0$).

2 Solve $\frac{dy}{dx} = 2xy$, given $y(0) = 5$.
$\frac{1}{y}dy = 2x\,dx$, so $\ln|y| = x^2 + C$. Using $y(0) = 5$: $C = \ln 5$. Solution: $y = 5e^{x^2}$.

Growth, decay, and applications

3 A population grows according to $\frac{dP}{dt} = 0.05P$, with $P(0) = 2000$.

a Solve for $P(t)$. $P(t) = 2000e^{0.05t}$.

b Find $P(20)$, to the nearest whole number. $P(20) = 2000e^{1} \approx 5437$.

4 Newton's law of cooling gives $\frac{dT}{dt} = -k(T - 20)$ for an object cooling in a $20°$C room. If $T(0) = 100$ and $T(5) = 60$, find k and hence $T(t)$.
Solving: $T(t) = 20 + 80e^{-kt}$. Using $T(5) = 60$: $40 = 80e^{-5k}$, so $e^{-5k} = \frac{1}{2}$, giving $k = \frac{\ln 2}{5} \approx 0.1386$. So $T(t) = 20 + 80e^{-0.1386t}$.

More separable equations

5 Solve $\frac{dy}{dx} = \frac{y}{x}$, given $y(1) = 4$.
$\frac{1}{y}dy = \frac{1}{x}dx$: $\ln|y| = \ln|x| + C$, so $y = Ax$ for constant A. Using $y(1) = 4$: $A = 4$: $y = 4x$.

6 Solve $\frac{dy}{dx} = y(1 - y)$ (logistic-type equation), given $y(0) = \frac{1}{2}$, using partial fractions $\frac{1}{y(1-y)} = \frac{1}{y} + \frac{1}{1-y}$.
$\int\left(\frac{1}{y} + \frac{1}{1-y}\right)dy = \int dx$: $\ln|y| - \ln|1 - y| = x + C$, i.e. $\ln\left|\frac{y}{1-y}\right| = x + C$. Using $y(0) = \frac{1}{2}$: $\frac{y}{1-y} = 1$ at $x = 0$, so $C = 0$: $\frac{y}{1-y} = e^x$, giving $y = \frac{e^x}{1+e^x}$.

Mixing and applied models

7 A tank has 100 L of pure water. Brine with concentration 2 kg/L flows in at 3 L/min, and the well-mixed solution drains at the same rate, so the volume stays constant. Let $y(t)$ be the amount of salt (kg) at time t. The differential equation is $\frac{dy}{dt} = 6 - \frac{3y}{100}$.

a Find the equilibrium value of y (where $\frac{dy}{dt} = 0$). $6 = \frac{3y}{100}$ gives $y = 200$ kg.

b Explain, in words, what this equilibrium represents physically. In the long run, the salt content approaches 200 kg, at which point the rate of salt entering equals the rate leaving, keeping the amount constant.

Slope fields and qualitative behavior

8 For the differential equation $\frac{dy}{dx} = x - y$, find the slope of the solution curve at the points $(0, 0)$, $(1, 0)$, and $(1, 1)$.

At $(0, 0)$: slope $= 0 - 0 = 0$. At $(1, 0)$: slope $= 1 - 0 = 1$. At $(1, 1)$: slope $= 1 - 1 = 0$.

9 Verify that $y = x - 1 + 2e^{-x}$ is a solution to $\frac{dy}{dx} = x - y$, by substituting into both sides.

$\frac{dy}{dx} = 1 - 2e^{-x}$. Right side: $x - y = x - (x - 1 + 2e^{-x}) = 1 - 2e^{-x}$. Both sides equal $1 - 2e^{-x}$, confirming the solution.